

Figure 5.3.7.4(a) Piston Pin

b) Piston Rings

Depending upon the horse power of the engine, one or more piston rings is used. It provides a tight seal between the piston and the cylinder wall thus preventing leakage of combustion gases. The performance of the engine is reduced, if the piston rings are worn out and it can be replaced. Piston rings are classified as

i) Compression ring

ii) Oil scrapper ring

Compression ring

The compression ring seals the combustion chamber from any leakage during the combustion process. When the air-fuel mixture is ignited, pressure from combustion gases is applied to the piston head, forcing the piston toward the crankshaft. The pressurized gases travel through the gap between the cylinder wall and the piston and into the piston ring groove. Combustion gas pressure forces the piston ring against the cylinder wall to form a seal. It also transfers heat from the piston to the cylinder wall. No. of compression ring is depending upon the compression ratio of the engine. Higher the compression ratio more the number of piston rings. Compression rings are made up of Cast Iron.

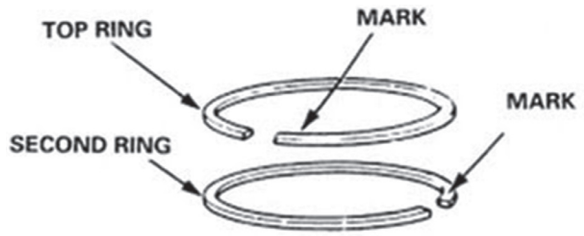


Figure 5.3.7.4(b)(i) Compression Rings

Oil scrapper ring

The bottom most ring is called as oil scrapper ring. The lubricating oils are sprinkled on the cylinder wall and inner side of the piston through the connecting rod oil passage. This ring is used to scrape excess lubricating oil from the cylinder walls, thus prevents the lubricating oil from getting into the combustion chamber of the cylinder Oil scrapper rings are made up of cast iron.

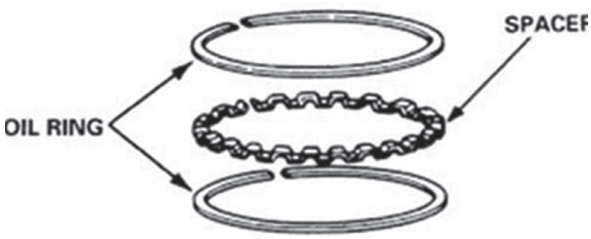


Figure 5.3.7.4(b)(ii) Oil Scrapper Ring

5.3.8 Connecting Rod

It interconnects the piston and the crankshaft and transmits the gas forces from the piston to the crankshaft. The two ends of the connecting rod are called as small end and the big end. Small end is connected to the piston by gudgeon pin and the big end is connected to the crankshaft by crank pin. Connecting rod is made up of Forged steel.

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